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Chapter · July 2011

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Current Market Trends and Future Directions

Caroline T. Yamaguishi, Michele R. Spier, Juliano De Dea Lindner,
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Abstract Probiotic products have been used worldwide and they are increasingly gaining popularity. Current trends in the consumption of probiotics are associated with increased levels of health-consciousness, and the availability of probiotics in the form of dietary supplements. Several companies have profited by marketing these products in different forms, with different purposes, and with recommendation

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for all ages. Important aspects in maintaining the viability and bioactivity of probiotic strains during processing and storage are also discussed in this chapter. The probiotic consumption by infants and the elderly has been supported by scientific evidences and represents a potential niche markets that are in developing and has led to the growth segment of probiotic products.

1 Introduction

Probiotics have achieved a prominent position in the global food market. The consumption of functional foods, such as probiotics, has heightened healthcare awareness and has led to better economic development of this market segment. Functional foods provide nutrients and promote health benefits; probiotics can be consumed in the daily diet and used as preventive medicine against diseases affecting all age groups.

Among the countries that have shown growth in the probiotic market, Europe represents the largest and fastest growing market, followed by Japan (Global Industry Analytic 2010). Currently, there is a wide range of probiotic products offered by companies such as BioGaia Biologics AB, Christian Hansen A/S, ConAgra Functional Foods, Danisco, Groupe Danone, Lifeway Foods, Inc., Nestle S.A., Seven Seas Ltd., Valio and Yakult Honsha Co. Ltd.

This market segment is established mainly in dairy products, but other products have shown relative importance (e.g., dietary supplements, probiotic baked products, probiotic ice creams, probiotic chocolates) leading to a diversification and popularisation of these products. The main claims of the probiotic products are based on several studies that support their potential use in the prevention of inflammatory diseases, allergies, cholesterol reduction, gastrointestinal disorders and some types of cancer.

2 Economic Relevance of the Probiotic Products

The main factors that promote growth in the global market of probiotics are increasing levels of health-consciousness and the availability of probiotics in the form of dietary supplements. The market of probiotic products generated US\$ 15.9 billion in 2008 and is forecast to reach US\$ 28.8 billion in 2015. The major factors that have facilitated market growth of probiotics are the appropriate components for formulation and the scientific knowledge of the provided benefits (Markets and Markets 2009; Global Industry Analytic 2010).

Probiotics are already used extensively in Europe and in some parts of Asia, and they are gaining in popularity worldwide, particularly in the United States. Europe represents the largest and fastest growing market for them. In addition,

Europe, Germany and the United Kingdom represent approximately 45% of the total market. Japan is the second largest market and is expected to grow at more moderate rates (Global Industry Analytic 2010).

More than 500 probiotic food and beverage products have been introduced in the past decade. These products have been commercialised on different levels, depending on their relevance for health. For instance, probiotic chocolates have garnered more market share than probiotic cheese and butter because probiotic chocolates offer more health benefits without fortification. The leading developers and suppliers of probiotic strains include Danisco, Morinaga and BioGaia. These products are used by companies such as Nestle and Attune. The United States market is growing rapidly due to the general affinity of the population towards probiotic dietary supplements. Probiotic products may include functional foods and beverages, dietary supplements, specialty nutrients and animal feed products. The most common applications are in regular consumption, probiotic therapy and disease prevention.

The main participants who leads the market for probiotics include BioGaia Biologics AB, Chr. Hansen A/S, ConAgra Functional Foods, Danisco, Groupe Danone, Institut Rosell, Lifeway Foods, Inc., Natren, Inc., Nestle S.A., Seven Seas Ltd., Stonyfield Farm and Yakult Honsha Co. Ltd. There is a lot companies which commercialises probiotic products such as Alpharma, Alltech Biotechnology, Amerifit Brands, Arla Foods, Attune Food Inc., Biogaia AB, Bomac Vets Plus, Calpis Company, Cargill, CD Pharma, China-Biotics Inc, CHR Hansen, Danisco, Danone, DSM Food Specialities, Ganeden, Garden of Life, General Mills, Jamieson Laboratories, Arrow Formulas, Kashi Company, Kirkman, Kraft Foods Inc., Lallemand Inc., Life Way Food Inc., Morinaga Milk Industry Co. Ltd, Mother Dairy, Muller Dairy Ltd, Natren, Nebraska Cultures, Nestle, Now Foods, Nutraceutix, Inc, Probi AB, Skanemejerier, Stonyfield Farm, Valio, Well's Dairy Inc., Wild Wood, Yakult Honsha, Yeo Valey, Yo Cream International Inc. The more recent idea has been based on fortifying fermented desserts with proteins, minerals and probiotic cultures (Markets and Markets 2009; Global Industry Analytic 2010).

3 Current-Commercially Available Probiotic Products

There are new probiotic products available in the market which can present several applications. Industries may use a mixture of strains in several products such as cereal bars, breads, dairy products, Asian fermented foods and some juices and soy beverages. They are customisable and stable, and the benefits have been proven.

Probiotics are available in foods, dietary supplements and in some other forms (e.g., capsules, tablets, pills and powders). The bacteria may have been present originally or added during preparation. Commercial probiotic products are available in liquid preparations and solid formulations (freeze-dried, spray-dried and pills). The market also has offered consumer probiotic products for animal feed.

3.1 Liquid Preparations

Probiotic preparation in liquid form is the most common product that is commercially available to consumers because these products originated and developed from yoghurt formulations. Several liquid formulations, commercial names, companies, probiotic cultures, compositions and applications are presented in Table 1.

3.2 Solid Probiotic Preparations

Different products are commercially available in solid form. The manufacturing of probiotics varies with purpose and product. The most commonly found probiotics are freeze-dried and spray-dried powders and pills, which may or may not be encapsulated. Still, probiotic strains can be incorporated into food matrices (Table 2) promoting the enrichment of the product in nutritional and market value.

3.2.1 Freeze-Dried and Spray-Dried Preparations

Probiotic preparations that are used for food applications are commonly supplied in dried form or are frozen as either freeze-dried (Table 3) or spray-dried powders. Materials used for coating are water-soluble polymers for spray-drying. Besides freeze-dried or spray-dried the material for coating may be waxes, fatty acids, water-soluble, water-insoluble polymers and monomers.

These processes have a high yield for application in several areas, but the exposure of the microorganism to high temperatures during spray-drying can be detrimental to the integrity of the live bacterial cells. The effect of spray-drying on the cell membrane can lead to increased cell permeability, which may result in leakage of intracellular components from the cell into the surrounding environment. The cytoplasmic membrane is one of the most susceptible sites to the stresses associated with spray-drying in bacterial cells; the cell wall, DNA and RNA are also known to be affected, leading to loss of metabolic activity (Teixeira et al. 1997).

To solve this problem a variety of protectants have been added to the drying media before freeze-drying or spray-drying to ensure the viability of probiotics during dehydration, including skim milk powder, whey protein, trehalose, glycerol, betaine, adonitol, sucrose, glucose, lactose and polymers such as dextran and polyethylene glycol (Hubalek 2003; Morgan et al. 2006). Growth-promoting factors including various probiotic/prebiotic combinations and granular starch have been shown to improve culture viability during drying and storage. Recently, the incorporation of soluble gum in a milk-based medium prior to spray-drying has been studied with probiotic cultures.

Table 1 Probiotic liquid products that are commercially available, probiotic cultures, composition and its applications

Product (company)	Probiotic culture	Composition	Application
Yakult® (Yakult Honsha Co. Ltd.)	<i>Lactobacillus casei</i> Shirota 10 ⁹ CFU/bottle, <i>Bifidus regularis</i> (<i>Bifidobacterium animalis</i>),	Water, sugar, skim milk powder, glucose, natural and artificial flavours	Probiotic beverage, intestinal flora reposition, improve digestion
Activia® Yogurt (Danon)	<i>L. bulgaricus</i> , <i>S. thermophilus</i>	Different variation (strawberry, natural, peaches, vanilla) Presents in the form of milk, buttermilk, yoghurts, fermented milks, milks, daily dose drinks, juices, berry soups, cheese and capsules	Help regulate digestive system
Gefilus® Valio Gefilus® and Gefilus® (FinFood)	<i>Lactobacillus</i> GG® (<i>L. rhamnosus</i> GG, ATCC 53103 or LGG) <i>Bacillus subtilis</i> var natto, <i>Bifidobacterium animalis</i> , <i>B. bifidum</i> ; <i>B. longum</i> , <i>L. acidophilus</i> <i>L. bulgaricus</i> , <i>L. casei</i> , <i>L. delbrueckii</i> , <i>L. fermentum</i> , <i>L. plantarum</i> , <i>Lactococcus lactis</i> , <i>L. lactis</i> subsp. <i>Diacetylactis</i> , <i>S. cerevisiae</i> , <i>Streptococcus thermophilus</i>	Purified water, sugar cane molasses, rock salt, sea salt, blueberry, cherry and pomegranate juice concentrates, Xtra Immunity™ brown rice extract, SCD Probiotics cultures	Improve digestion
SCD Essential Probiotics™ (Sustainable Community Development)			Regulate digestion and supporting personal wellness
SVELTY® Gastro Protect (Nestle)	<i>Lactobacillus johnsonii</i> La1	A fermented drink milk, flavour, sugars	Controls <i>H. pylori</i> infection and stomach discomfort
LC1 Yoghurt® (Nestle)	<i>L. johnsonii</i> La1 and acidophilus bacteria	A probiotic yoghurt, fermented milk, flavours, sugars	Regulates digestion, protection against pathogens
Actimel® (Danon)	<i>L. casei</i> (<i>Defensis</i>)	Milk, sugar, flavours	Protection against pathogen

Available at online database of each manufacturer

Table 2 Probiotic solid products that are commercially available, probiotic cultures, composition and its applications

Product	Probiotic culture	Composition
Probiotic Chewy Cereal Bars	Ganeden BC30	5 g fibre, 2 g protein, prebiotics, omega-3 fatty acids, total 110 cal
Chocolate Probiotic Bars	<i>Lactobacillus acidophilus</i> , <i>L. casei</i> , <i>Bifidobacterium lactis</i>	Milk Chocolate Coating (evaporated cane juice, chocolate, cocoa butter, inulin, non-fat milk, calcium carbonate, anhydrous milk fat, soy lecithin, vanilla), organic brown rice crisps (organic brown rice flour, organic molasses, calcium carbonate)
Chocolate Probiotic Bars Chocolate Crisp® ATTUNE		Dark Chocolate (unsweetened chocolate, sugar, cocoa powder, lecithin, vanilla extract), antioxidant blend (natural cocoa, açai, blueberry powders) and probiotics
XoBiotic™ squares (MXI Corp.™)	<i>L. helveticus</i> R0052 and <i>B. longum</i> R0175	
Heini's Yogurt Cultured Cheese (Bunker Hill Cheese Company)	<i>L. acidophilus</i> , <i>L. casei</i> , <i>B. lactis</i>	Milk, yoghurt cultures, coagulants, probiotic cultures and salt

Database from companies' websites

Table 3 Freeze-dried products

Product	Probiotic culture	Application
FloraFIT® (Danisco A/S)	<i>Bifidobacterium breve</i> Bb-03, <i>B. lactis</i> Bi-07, <i>B. lactis</i> Bl-04, <i>B. longum</i> Bl-05, <i>Lactobacillus acidophilus</i> La-14, <i>Lactobacillus bulgaricus</i> Lb-64, <i>L. Brevis</i> Lbr-35, <i>L. casei</i> Lc-11, <i>Lactococcus lactis</i> Ll-23, <i>L. plantarum</i> Lp-115, <i>L. paracasei</i> Lpc-37, <i>L. rhamnosus</i> Lr-32, <i>L. salivarius</i> Ls-33, <i>Streptococcus thermophilus</i> St-21	Food and beverages
HOWARU® Premium Probiotics (Danisco A/S)	<i>L. acidophilus</i> NCFM™	A probiotic product that can be applied in beverages, confectionery, dairy, dietary supplements and frozen desserts
Yógourmet Products (Lyo-San, Inc.)	<i>L. casei</i> , <i>B. Bifidus</i> , <i>L. acidophilus</i>	Starters for yoghurt manufacture
Biorich® (Chr. Hansen A/S)	<i>L. acidophilus</i> LA-5 and <i>Bifidobacterium</i> BB-12	Starters for yoghurt manufacture

Database from companies' websites

3.2.2 Pills

Some probiotic pills that are commercially available (Table 4) contain probiotic strains and prebiotics, which help with the growth and colonisation of good bacteria within the intestinal flora.

Table 4 Probiotic pills commercially available

Product	Probiotic culture	Composition	Benefits
Nutraelle® DigestiveCare	<i>Lactobacillus acidophilus</i> : 5×10^9 CFU/pill. <i>Bifidobacterium</i> <i>bifidum</i> : 5×10^9 CFU/ pill	Prebiotic (inulin Powder (Chicory Root) – Prebiotic	Body benefits
	10^9 live bacteria per capsule of <i>Bifidobacterium</i> <i>infantis</i> 35624 found exclusively in Align		
Align®		Only <i>B. Infantis</i> 32.5 mg of lactose monohydrate (not derived from milk), 2.85 mg of Mg distearate and titanium dioxide	Help in the restoration of the digestive system
Florastor®	<i>Saccharomyces boulardii</i> lyo (250 mg or 5 billion live cells)	E171 used to distribute the yeast evenly	Promote intestinal health and maintain normal bowel function
			Promote good digestion, support immune system and restore the natural intestinal balance
Culturelle®	<i>Lactobacillus rhamnosus</i> GG (10^9 billion cells) <i>B. bifidum</i> 7.5×10^9 ; <i>L. acidophilus</i> 3×10^9 ; <i>L. rhamnosus</i> 1.35×10^9 ; <i>B. breve</i> 75×10^7 ; <i>B. longum</i> 75×10^7 ; <i>L. casei</i> 75×10^7 ; <i>L. plantarum</i> 45×10^7 ; <i>L. lactis</i> 3×10^8 ; <i>L. bulgaricus</i> 75×10^6 ; <i>L. salivarius</i> 15×10^6	Inulin	Provides probiotic support for the small intestine, which produces enzymes to help digest foods
ReNew Life Ultimate Flora Adult® (Wellness Nutrition)	<i>L. rhamnosus</i> 8.5×10^9 , <i>L. casei</i> 5.87×10^9 , <i>L. plantarum</i> 3.75×10^9 , <i>L. acidophilus</i> 2.5×10^9 , <i>L. brevis</i> 1.25×10^9 , <i>B. lactis</i> 1.25×10^9 , <i>B. longum</i> 1.25×10^9 , <i>L. paracasei</i> 25×10^7 , <i>L. salivarius</i> 25×10^7 , <i>L. bulgaricus</i> 125×10^6 , FOS (fructooligosaccharide), Total Probiotic Cultures 125×10^9		
Ultimate Flora Women's Care® 25 Billion (Renew Life Formula's, Inc.)		FOS, vegetable capsule (vegetable fibre, water)	Application: restores a healthy yeast balance; promotes vaginal & urinary tract health; helps replenish healthy vaginal flora levels; strengthens natural defences

(continued)

Table 4 (continued)

Product	Probiotic culture	Composition	Benefits
Rapid Yeasts Relief [®] (part I and II) (Renew Life)	Part I (<i>L. acidophilus</i> , <i>L. rhamnosus</i> , <i>L. reuteri</i> , <i>L. paracasei</i> , <i>L. salivarius</i> , FOS) Total probiotic cultures 20 × 10 ⁹ CFU/capsule	Part II (Enzyme blend containing protease, papain, cellulase, hemicellulase, lysozyme, lactoferrin, amylase, invertase, beta glucanase, malt diastase, glucoamylase and FOS	Against yeast infection
Flora Bear [®] for Kids Renew Life	Probiotic blend (<i>L. acidophilus</i> , <i>B. bifidum</i> , <i>B. infantis</i>). Total 1 × 10 ⁹ CFU/tablet	FOS 100 mg, L-glutamina 100 mg, sugars 1 g/tablet. Natural orange flavour, citric acid, vegetable stearate and vegetable stearine	Helps digestion
Buddy Bear [®] Probiotic	<i>L. acidophilus</i> 9 × 10 ⁸ CFU, <i>B. bifidus</i> 50 × 10 ⁶ CFU, <i>B. infantis</i> 50 × 10 ⁶ CFU	FOS 100 mg, organic cane extract, Orange fruit crystals, natural orange flavour, citric acid, vegetable stearate	Improves intestinal digestion
INNEOV Solaire skin probiotic (Nestle and L'Oreal,)	Strain not reported	Licopene, b-carotene Boneset (Eupatorium perfoliatum, Listletoe Leaf (Viscum album), cellulose, lactose, maltodextrina	Capsules, helps to improve skin protection against UV-rays Prevents yeast infection and helps to relieve the symptoms caused by infection in women
AZO Yeast	<i>Lactobacillus sporogenes</i>		Drops used to reduce colic; improves digestive health and function and boosts immunity
BIOGAIA [®] AB probiotics drops (Biogaia)	<i>Lactobacillus reuteri</i> 10 ⁸ FCU		

Database from companies' websites

3.3 Products for Feed

Animal feed companies and researchers have been looking for alternative products and strategies that can help maintain animal gut health by preventing or reducing

Table 5 Probiotic product for feed

Product	Probiotic culture	Application	Company
BioPlus 2B	<i>Bacillus subtilis</i> , <i>B. licheniformis</i> <i>Lactobacillus acidophilus</i> , <i>L. casei</i> , <i>Bifidobacterium</i> <i>bifidum</i> , <i>Enterococcus</i> <i>faecium</i> , contains rice mill by-product, calcium carbonate	Growth promoter for pigs and broilers	Chr. Hansen
Primalac	<i>Saccharomyces cerevisiae</i> 1026	Poultry, beef and aquaculture (fish)	Star Labs
Yea-Sacc® 1026	<i>Lactobacillus acidophilus</i> , <i>L. plantarum</i> , <i>L. casei</i> , <i>Enterococcus faecium</i>	Dairy, beef, equine, pigs, sheep	Alltech
All-Pro Biotic™		Cattle, swine, sheep, goats, poultry, horses Cattle and other ruminants. Animals when birthing, weaning, vaccinating, handling, undergoing changes in weather, shipping or in antibiotic treatment	Bomac Vets Plus sp.
Power Punch™	Lactic acid bacteria DDS-1	Animals during shipping, adverse weather, overcrowding or ratio changes, all of which can trigger a stress reaction in the animal.	Bomac Vets Plus sp.
Probiotic Paste™	Lactic acid bacteria DDS-1, contains inulin		Bomac Vets Plus sp.

Database from companies' websites

damages caused by attacks from pathogens. An alternative and effective approach to antibiotic administration in livestock is the use of probiotics, which can help improve gut microbial balance and, therefore, the natural defence of the animal against pathogenic bacteria (Modesto et al. 2009; Patterson and Burkholder 2003). In recent years, there has been considerable interest in using probiotic microorganisms and organic acids as alternatives to the use of antibiotics in feed. Table 5 shows some examples of products that contain probiotic cultures for application in feed.

4 Ongoing Innovations: Improvement of Probiotic Viability and Bioactivity

Technological characteristics such as probiotic viability, bioactivity and safety must be taken into consideration during the selection process of probiotic microorganisms. Functional aspects include the following: viability and persistence in the gastrointestinal tract, immunomodulation and antagonistic and antimutagenic properties. Probiotic strains must first be able to be manufactured under industrial

conditions. Furthermore, they have to survive and retain their functionality during storage and in the foods into which they are incorporated, all without producing off-flavours. Factors related to the technological and sensory aspects of probiotic food production are important; the food industry can only succeed in promoting the consumption of functional probiotic products in the future by satisfying the demands of the consumer.

Good viability and activity of probiotics are considered prerequisites for optimal functionality. It may be sufficient for certain probiotic strains to grow well during the initial production steps (to obtain high enough numbers in the product), but they do not necessarily retain good viability during storage.

The product carrier, the conditions of processing and storage and the use of techniques that protect the probiotic microorganisms from harsh environments are several variables that tend to influence the viability and stability of the strains, and some aspects of these factors are described below.

4.1 Selection of Suitable Product Carrier

In products such as yoghurt, acidity of the product (pH), storage temperature and storage time influence the viability of probiotic bacteria. The storage temperature at 4°C, thus reflecting refrigerator conditions, was the most important factor in maintaining the viability of the probiotic bifidobacteria during the 4-week storage period. Room temperature is the most damaging factor in storing probiotic bacteria. Suitable product carriers can be based on formulations that require refrigeration temperature (Technical Research Centre of Finland 2003).

Food is a good carrier to deliver probiotic bacteria. Milk is a favourable medium for the growth of lactic acid bacteria (e.g., *Lactobacillus* and *Bifidobacterium*) because it contains lactose, which is easily metabolised by these bacteria. However, concern with the choice of an appropriate product carrier has been less relevant due to the availability of highly concentrated commercial formulations (e.g., freeze-dried preparations). The suitable carrier should not be dairy-based product necessarily, but it should have the ability to deliver high viable populations.

4.2 Improvement in Culture Stability

During the steps of processing, food manufacturing, transport and storage, probiotic strains are susceptible to conditions of heat, humidity or exposure to oxygen. Most research to date has been on the physiological effects and stability of probiotics in the gut. However, researchers have developed a protection system combining processing technology and a mix of nutrients that protect probiotics during processing, transport and storage. This protection system is applied in a variety of products, including powders.

Unlike most probiotic strains currently found in popular brands of yoghurt, some probiotics that have been developed are able to survive the heat and pressure of manufacturing processes and the acidic stomach environment. These new probiotic strains also remain viable without refrigeration, making it ideal for inclusion in shelf-stable products such as cereal bars, biscuits, breads, cereals formulations. One example is the Sunny Crunch bar, a new product containing the probiotic strain GanedenBC30 (Ganeden Biotech Co. Mayfield Heights, Ohio, USA), which can remain viable without refrigeration; this makes it ideal for inclusion in products stored at room temperature.

There are several strains of probiotic bacteria, and their ability to survive through different stress factors is highly variable. One alternative for probiotic bacteria might be protection through combining them with growth-promoting substances. These so-called “prebiotics” can support the survival of certain strains, but finding the suitable probiotic and prebiotic pairs requires further study (Technical Research Centre of Finland 2003).

When spray-drying is used for the preservation of potential probiotic cultures, much of their activity is typically lost after a few weeks of storage at room temperature. This is associated with stress that is induced by temperature changes, phase changes and drying, a combination of which tends to damage cell membranes and associated proteins. The use of protectant substances can improve culture viability during the drying and storage steps.

One of the greatest drawbacks associated with the use of dietary cultures in fermented milk products is the lack of acid tolerance of some species and strains. “Overacidification” or “post-acidification” is mainly due to the uncontrolled growth of strains such as *L. bulgaricus* at low pH values and refrigerated temperatures. The effect of pH on the viability of some microorganisms has been investigated by some researchers. Studies show that the pH must be maintained above 4.6 to prevent the decline of bifidobacteria populations. Also, *L. casei* cultures seem to be more adaptable to the acidic environment.

The use of probiotic cultures as an additive to cheese is another point to be considered. The effect of different concentrations of NaCl on the viability of some microbial strains has been evaluated, and the results show that there is an opposite relationship between the viable counts of all microorganisms and the salt concentration along the storage period. Therefore, salt-tolerant strains can be used as a criterion in selecting probiotic bacteria for cheese manufacturing.

During the manufacture of semi-hard and hard cheese, whey is entrapped in the curd particles and is expelled by the scalding step, which is performed at temperature ranging from 41 to 55°C. It was noted that the temperatures under 50°C had no significant changes in viability of various probiotics tested. By the way, at 55°C the viability of the non-microencapsulated species decreased significantly (Tamime 1993). Therefore, microencapsulation may have potential for increasing the survival of probiotics during the scalding of cheeses.

In addition, high concentrations of sugar added to milk before fermentation may result in conditions that are inhibitory to the growth of probiotics in yoghurt and other products; this will lead to long fermentation times and poor acidity

development. This fact may occur due to the adverse osmotic effects of the solutes in the product. Shah and Ravula (2000) demonstrated that the addition of sugar may be deleterious to the growth of probiotic bacteria especially in products such as fermented frozen dairy desserts, which contain approximately 16% sugar. Walker (2001) also showed that yeast strains are more osmotolerant than bacterial strains due to the elasticity of yeast cell walls; yeast cell walls also buffer against water loss.

4.3 *Microencapsulation*

The survival of probiotic bacteria in different matrices is a constant challenge. Many studies have demonstrated low viability in products such as yoghurts, fermented milks and dairy-fermented desserts. Microencapsulation techniques can provide protection for probiotics through the entrapment of cells in hydrocolloid beads, increasing the viability in food matrices and stability during storage life and protecting against the adverse conditions of the gastrointestinal tract.

Microencapsulation can be employed by several techniques, including spray-drying, freeze-drying and fluidised bed-drying, which converts the cultures to a concentrated powdered form (Krasaekoopt et al. 2003). These techniques allow for the release of bacteria in the product.

Considering the method used to form the beads, microencapsulation techniques can be divided into two groups: extrusion (droplet method) and emulsion or two-phase system (Fig. 1). Using these methods, the survival of probiotic bacteria can be increased by up to 70–95% (Heidebach et al. 2009; Pimentel-González et al. 2009).

4.3.1 **Extrusion**

The supporting material used in extrusion is alginate. It is a natural polymer, also known as alginic acid, that is extracted from seaweed and composed of various proportions of 1–4 linked β -D-mannuronic (M) and α -L-guluronic (G) acids, which may vary in composition and sequence depending on the source. The binding of divalent cations (e.g., Ca^{+2}) and subsequent gel formation depend on the composition and arrangement of the blocks of alginic acid residues (Gemeiner et al. 1994). The droplets may be formed when sodium alginate solution is added to a calcium solution with an instantaneous polymerisation (precipitation) of calcium alginate followed by a gradual gelation of the interior as cross-linked calcium ions permeate the cells. Among the advantages of this support material are its cheapness, simplicity and biocompatibility (Martinsen et al. 1989). In general, authors have used a concentration of sodium alginate ranging from 1 to 2% and 0.05 to 1.5 M CaCl_2 (Krasaekoopt et al. 2003). The increasing concentration or viscosity of sodium alginate leads to a decrease in bead size. In addition, the composition of the alginate

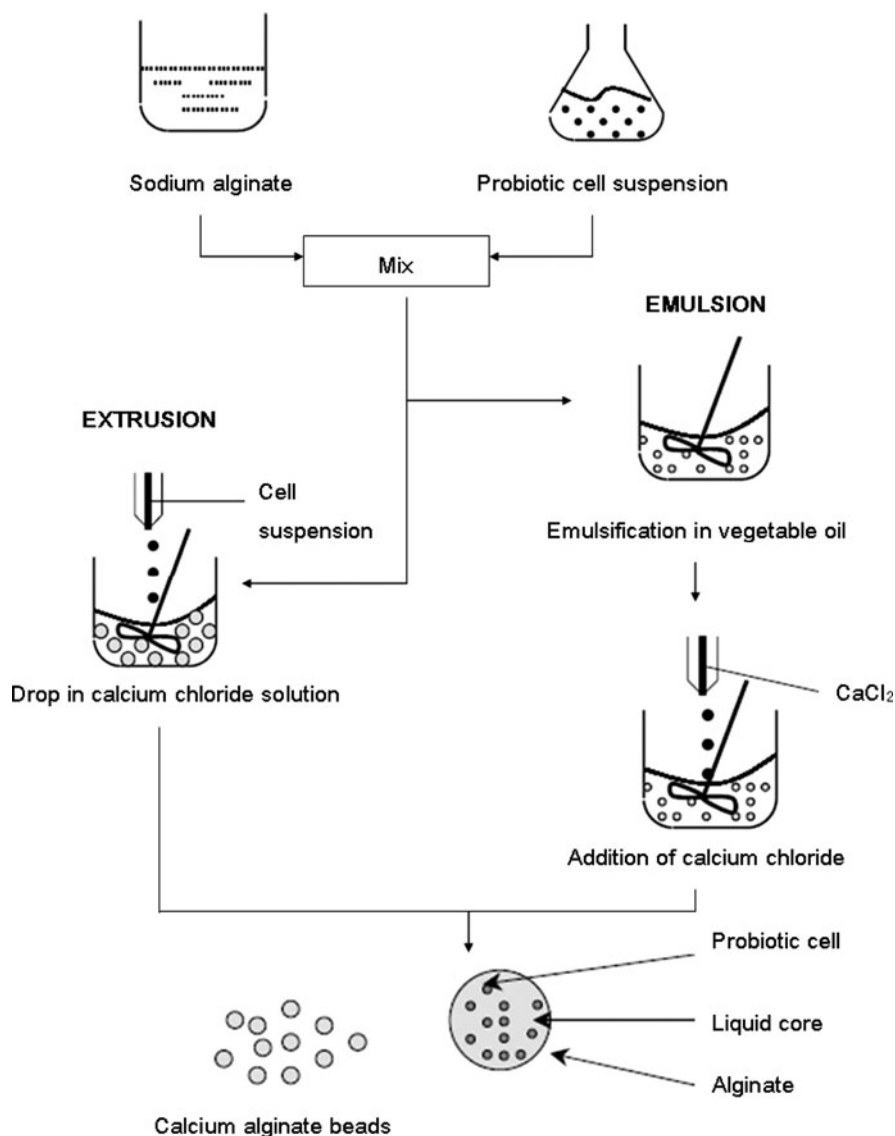


Fig. 1 Microencapsulation of probiotic bacteria in sodium alginate by extrusion and emulsion techniques. Based on Krasaekoopt et al. (2003)

may influence the bead size (e.g., “low guluronic” alginates result in small beads) (Martinsen et al. 1989).

In addition, this method presents some disadvantages because it can be difficult to scale up, and the speed of gel formation is slow compared to the speed of the emulsion technique.

4.3.2 Emulsion

In this method, a small volume of cell polymer suspension (discontinuous phase) is added to a large volume of vegetable oil (continuous phase), which could be soybean, sunflower, canola or corn oil. The mixture is homogenised, and there is formation of a water-in-oil-emulsion. Afterwards, the water-soluble polymer is insolubilised (cross-linked) to form gel beads within the oil phase (Fig. 1). The beads can be harvested later by filtration. The bead size depends on the speed of agitation and varies from 25 to 2 mm (Krasaekoopt et al. 2003).

Other emulsifiers can also be used as the continuous phase (e.g., Tween 80, white light paraffin oil and mineral oil). A mixture of κ -carrageenan and locust bean gum, cellulose acetate phthalate (CAP), alginate, gelatin and chitosan (Anal and Singh 2007) can be used as supporting material in microencapsulation by the emulsion technique.

This method is relatively new for the food industry, and it is easy to scale up aside from providing encapsulation and entrapment of core materials. The bead size is smaller (25–2 mm) than beads formed in the extrusion technique (2–5 mm). However, the cost of using the emulsion technique can be higher due to the use of vegetable oil (Krasaekoopt et al. 2003).

4.3.3 Enhancement of Survival by Microencapsulation Techniques

Recent advances have been achieved with the encapsulation of probiotic cultures. There are several systems employed, and they have been successful because encapsulated bacteria have better survival rates than non-encapsulated bacterial cells, even in simulated gastric conditions. Nevertheless, in some situations it is necessary to cover the beads and proceed with encapsulation with polymer. In general, measures such as cross-linking with cationic polymers (polyethyleneimine, polypropyleneimine and glutaraldehyde), coating with other polymers (chitosan), mixing with starch and using additives (glycerol) help to increase the stability of beads.

The use of entrapped probiotic microorganisms has been employed in the production of dairy products (e.g., yoghurt, cheese and frozen milk products) and biomass production. Encapsulated microorganisms present advantages such as constant characteristics, decrease of incubation time, survival in harsh conditions (e.g., freezing temperatures and acidic environments), protection against attack by bacteriophages, improved stability during storage and productivity.

There are many polymers currently used in the encapsulation of cells that are of non-dairy origin, which limits their use in dairy products. Also, they result in low-density gels with relatively large bead production (1–3 mm) (Heidebach et al. 2009). Furthermore, they can change the sensorial quality of the enriched products. As an alternative to all of these drawbacks, milk proteins have been proposed for use as encapsulant agents because they offer appropriate physico-chemical properties for food applications.

A novel microencapsulation process of probiotic cells that is based on emulsification and rennet-induced gelation of skim milk concentrates has been developed by Heidebach et al. (2009). The researchers noted a high encapsulation yield by this method for *L. paracasei* F19 (105%) and *B. lactis* Bb12 (115%). Moreover, when tested in simulated gastric conditions (pH 2.5 at 37°C for 90 min), both encapsulated strains presented higher viable cell numbers than the free cells; this result was despite the influence of encapsulation and was more pronounced for *B. lactis* Bb12 than for *L. paracasei* F19 (2.8 log units CFU.g⁻¹ and 0.8 log unit CFU.g⁻¹ higher compared to free cells, respectively).

The efficiency of encapsulation can be enhanced by employing prebiotics as coating materials. Chen et al. (2005) optimised proportions of sodium alginate (1% w/v), peptide (1% w/w) and fructooligosaccharides (3% w/w) as coating materials for encapsulating cells of *L. acidophilus*, *L. casei*, *B. bifidum* and *B. longum*. They observed a high survival rate of microencapsulated cells compared to free cells after storage in refrigerated milk for 2 weeks. The use of prebiotics was found to be beneficial because probiotic microcapsules presented counts of 10⁵–10⁶ CFU/g in contrast to the 10²–10³ CFU/g count of free cells after storage for 12 weeks. However, there are reports that prebiotics can also be detrimental to the encapsulation yield (Chávarri et al. 2010).

5 Future Larger Markets: Infants and the Elderly

The trends of the market have led probiotic companies to develop formulations and products directed to specific target populations such as infants, elderly, athletes and immunocompromised patients. Some studies have demonstrated positive effects when probiotics are administered orally in in vivo trials. Investment in this sector has proven interesting, and the main appeal is using these products as preventive measures against various types of diseases by enhancing the defence system, which decreases the susceptibility of these populations during more aggressive treatments. An overview is presented below regarding the relevance and marketing of probiotic products for infants and the elderly.

5.1 Probiotic for Infants and Children

The first bacterial source for newborns is through contact with the mother. The intestine is colonised first by facultative Gram-positive cocci (staphylococci, streptococci and enterococci) and enterobacteria. It is possible that anaerobic bacteria start to colonise on the second day (Harmsen et al. 2000). The genera found in the intestinal microbiota are *Bacteroides*, *Lactobacillus*, *Clostridium*, *Fusobacterium*, *Bifidobacterium*, *Eubacterium*, *Peptococcus*, *Peptostreptococcus*, *Streptococcus* and *Veillonella* (Savage 2005).

In addition, the composition of the indigenous intestinal microbiota of the infants may be influenced by factors such as mode of birth and breast feeding. In preterm infants, for example, the flora is characterised by the presence of coliforms, *Enterococcus* and *Bacteroides* species, where bifidobacteria are scarce (Savage 2005). Breast milk contains substances such as peptides and oligosaccharides, which promote the growth of bifidobacteria (Liepke et al. 2002). The intestinal bifidobacteria species that are influenced by breast feeding are *Bifidobacterium breve*, *Bifidobacterium infantis* and *Bifidobacterium longum*, whereas *Bifidobacterium adolescentis* are more common in the coming years of the children's development (Rautava 2007).

Studies have assessed the efficacy and safety of probiotics in neonates, but due to large differences between studies (e.g., probiotic strains used, age of the target population, dosage of the probiotic product, duration of probiotic supplementation), the results appear to be inconclusive. Nevertheless, many studies have been published suggesting the beneficial action of probiotics. Generally, probiotics provide resistance against colonisation by pathogenic bacteria through competition for nutrients or adhesion sites, production of antimicrobial substances or modulation of immune system. However, the action of probiotics must be strain-specific.

Some mechanisms may be related to the reduction of morbidity during infancy. Probiotics can promote intestinal barrier function, inhibit pathogen adherence and colonisation, degrade allergens, enhance immune maturation through the induction of IgA production and induction of regulatory T cells (Rautava 2007).

The use of probiotics in infancy has been most suggested in diarrhoeal cases. In this sense, *Lactobacillus* GG was shown to be effective in the prevention of acute infantile diarrhoea, including nosocomial spread of infection (Szajewska et al. 2001). Also, nosocomial spread of gastroenteritis and diarrhoea was prevented by bifidobacteria strains (Saavedra et al. 1994; Chouraqui et al. 2004).

Necrotising enterocolitis (NEC) is a gastrointestinal disease that affects mainly premature infants. Trials in animal models have showed that *B. infantis* reduces the risk of NEC in newborn rats (Caplan et al. 1999). In further studies, very low birth weight infants received supplementation based on combinations of probiotics (*L. acidophilus*, *B. infantis*, *S. thermophilus*, *B. bifidus*) that resulted in the reduction of the incidence of NEC (Lin et al. 2005; Bin-Nun et al. 2005).

In the first year of life, children can manifest atopic diseases (food allergy, atopic eczema, allergic rhinoconjunctivitis and atopic asthma), which cause discomfort and may even pose a risk to the infant's life, depending on the degree of inflammation. According to Romagnani (2004), the immune pathology of atopic diseases is characterised by T helper (Th) 2-driven inflammatory responsiveness against environmental conditions or dietary allergens. Probiotics can help reduce the risk of developing this type of disease. Rautava et al. (2002) tested the use of probiotics in breastfeeding mothers and neonatal infants with a high risk of developing atopic disease. Mothers received the supplementation with *Lactobacillus* GG 2–4 weeks before parturition. After birth, breastfeeding mothers and infants consumed probiotics for 6 months. In the group that was administered probiotics, a significant reduction of atopic eczema was found during the first 2 years of life. Interestingly, it

Table 6 Current probiotic products for infants and children

Mark product	Company	Probiotic strain	Health claims
BioGaia® Probiotic Drops	BioGaia®	<i>L. reuteri</i>	Protects the gastrointestinal tract, enhances immunity and improves colic symptoms
BioGaia® Probiotic Chewable Tablets	BioGaia®	<i>L. reuteri</i>	Promotes gut health and enhances immunity
Good Start® Probiotic	Nestlé®	<i>B. lactis</i>	Promotes healthy tract flora Assists the development of the digestive system, promotes a healthy digestive tract
Baby Cereals	Nestlé®	<i>B. lactis</i>	Increases intestinal bifidobacteria and promotes a healthy digestive tract
A.B. Pre & Pro	Bio-LiFE	<i>L. acidophilus</i> LA-5 and <i>B. lactis</i> BB-12	Reduction of allergic reactions, supports immunity, acts as an antibiotic treatment and prevents diarrhoea
Dicoflor 30	Vitis Pharma	<i>L. rhamnosus</i> GG	
OptiBac Probiotics For your Child's health	OptiBac® Probiotics	<i>L. acidophilus</i> Rosell-52, <i>B. infantis</i> Rosell-33 and <i>B. bifidum</i> Rosell-51	Promotes good digestive health and supports natural immune defences
EvoraKids™	Oragenics, Inc.	Probiora 3® (<i>S. oralis</i> , <i>S. uberis</i> and <i>S. rattus</i>)	Supports gum and tooth health by maintaining the oral health
Babio (biomilk with probiotics)	Goldim	BifiComplex® (<i>B. breve</i> , <i>B. longum</i> , <i>B. bifidum</i> and <i>B. infantis</i>)	Enhances the immune system
Happybaby	Nurture, Inc.	<i>L. acidophilus</i> , <i>B. lactis</i> , <i>L. casei</i> , <i>L. rhamnosus</i>	Protects against the development of allergies

Available at the online database of each manufacturer

was observed that the protective effect was most prominent in infants whose mothers received the probiotic supplementation.

According to Scientific Committee on Food of the European Commission, only probiotic bacteria that has been identified molecularly and has been found to have genetic stability can be added to formulae and infant foods. The stated level of viable bacteria should be 10^6 – 10^8 colony forming units (CFU) per gram of the formulation upon consumption.

Table 6 shows the commercialised probiotic products for infants and children.

5.2 Probiotic for the Elderly

As humans age, the immune system tends to fail (known as immunosenescence). This leads to a higher frequency of illnesses among the elderly. Dietary supplementation

with probiotics can enhance the immunity of the elderly, and they can be administered during vaccination and following surgical intervention.

This method of boosting immunity was demonstrated by Gill and Rutherford (2001), where 13 healthy elderly volunteers (age range 62–77 years) received a supplement with *Lactobacillus rhamnosus* HN001 (DR20™) for 3 weeks. Afterwards, the elderly subjects had enhanced phagocytic activity of peripheral blood polymorphonuclear (PMN) cells and monocytes (average of 35%).

The seasonal epidemic caused by the influenza virus represents one of the greatest threats to the elderly. Therefore, the use of probiotics can serve as a preventive measure against infection caused by the influenza virus because the clinical efficacy of the vaccine is lower in the elderly. For young healthy adults, the influenza vaccine provides clinical protection in 70–90% of cases; in the elderly, this protection is found in only 17–53% of cases. This may be related to the influenza vaccination seroconversion and seroprotection rates, which are two- to fourfold lower in the elderly when compared to the rates in younger people (Goodwin et al. 2006). Studies performed by Boge et al. (2009) showed that the consumption of two bottles containing 100 g per day of the commercial probiotic drink Actimel® (containing *L. casei* DN-114, combined with *Streptococcus thermophilus* and *Lactobacillus bulgaricus*) helps to increase specific antibody titres after vaccination against influenza in elderly over 70 years of age. In the pilot and confirmatory study, the seroconversion rates were higher in the probiotic group when compared to the rates in the control group for three strains tested (H1N1, H3N2 and B). Moreover, antibody levels in the probiotic group remained higher up to 9 weeks after vaccination.

Nowadays, some probiotics are targeted for the elderly and contain *Lactobacillus* and/or *Bifidobacterium*, mineral and other components. Just as aging and health factors change bone density, they also can alter the body's microbial content. Using probiotic dairy products that contain these strains can help restore and improve the intestinal flora balance, leading to a variety of associated health benefits, including mineral supplementation (Ca, Zn, Mg). These products include beverages such as milk and soymilk, drinkable yoghurts and other solid probiotic products (e.g., breakfast cereals, baked goods, snack foods, breakfast bars and crackers). An example is the product PuroFIT® calcium (FloraFIT company).

5.3 Safety Aspects

There are more safety concerns regarding side effects and bacteraemia (presence of bacteria in the blood) when considering the commercialisation of probiotic products, especially for children and elderly. In recent years, bacteraemia cases have not been reported. In addition, clinical trials have demonstrated the safety of bifidobacteria and lactobacilli, such as *L. casei* in infants and children (Borriello et al. 2003; Srinivasan et al. 2006). Although most of the published studies have not been designed to assess the safety of probiotics. However, competent organisations

(e.g., European Food Safety Authority – EFSA) have performed studies, which legally regulate the use of probiotic microorganisms.

6 Concluding Remarks

Probiotics can have a high potential for combating common diseases that afflict specific target populations. This field requires more studies to demonstrate potential adverse effects in neonates and infants. Also, it is important to consider that some properties (e.g., immunomodulation) appear to be strain-specific. More studies on the mechanisms of action of probiotics on the molecular level may indicate target strains (or combinations of strains) for specific functions and optimal dosages for each treatment. In people with genetic predispositions for developing certain diseases, prevention can be accomplished by administering probiotics daily. The discovery of new probiotic strains or engineered strains and the applications in other areas are the trends in probiotic research.

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